

Algorithm: Placement Management System

Step 1: Start the Program

1. Initialize three ArrayLists:
 - Student List
 - Company List
 - Placement List
2. Display the main menu.

Step 2: Add Student

1. Read Student ID.
2. Read Student Name.
3. Read Department.
4. Read CGPA.
5. Create a Student object.
6. Add the Student object to the Student List.
7. Display success message.

Step 3: Add Company

1. Read Company ID.
2. Read Company Name.
3. Read Job Role.
4. Read Minimum CGPA Requirement.
5. Create a Company object.
6. Add the Company object to the Company List.
7. Display success message.

Step 4: View Students

1. Check whether the Student List is empty.
2. If empty, display "No Students Available".
3. Otherwise, display all student details.

Step 5: View Companies

1. Check whether the Company List is empty.
2. If empty, display "No Companies Available".
3. Otherwise, display all company details.

Step 6: Place Student

1. Read Student ID.

2. Read Company ID.
3. Search for the student in the Student List.
4. Search for the company in the Company List.
5. If student is not found, display "Student Not Found".
6. If company is not found, display "Company Not Found".
7. Check whether the student is already placed.
8. If placed, display "Student Already Placed".
9. Compare student's CGPA with company's minimum CGPA.
10. If CGPA is less than required, display "Student Not Eligible".
11. Otherwise:
 - Mark student as placed.
 - Create a Placement object.
 - Add it to the Placement List.
12. Display "Placement Successful".

Step 7: View Placements

1. Check whether the Placement List is empty.
2. If empty, display "No Placements Yet".
3. Otherwise, display all placement records.

Step 8: Search Student

1. Read Student ID.
2. Search the Student List.
3. If found, display student details.
4. Otherwise, display "Student Not Found".

Step 9: Delete Student

1. Read Student ID.
2. Search the Student List.
3. If found, remove the student from the list.
4. Display "Student Deleted Successfully".
5. Otherwise, display "Student Not Found".

Step 10: Exit

1. Display "Thank You!".
2. Terminate the program.

Step 11: End